Statement Flow Control

- In python program statement may execute in a sequence, selectively or iteratively. Python programming support 3 Control Flow statements:

1. Sequence
2. Selection
3. Iteration

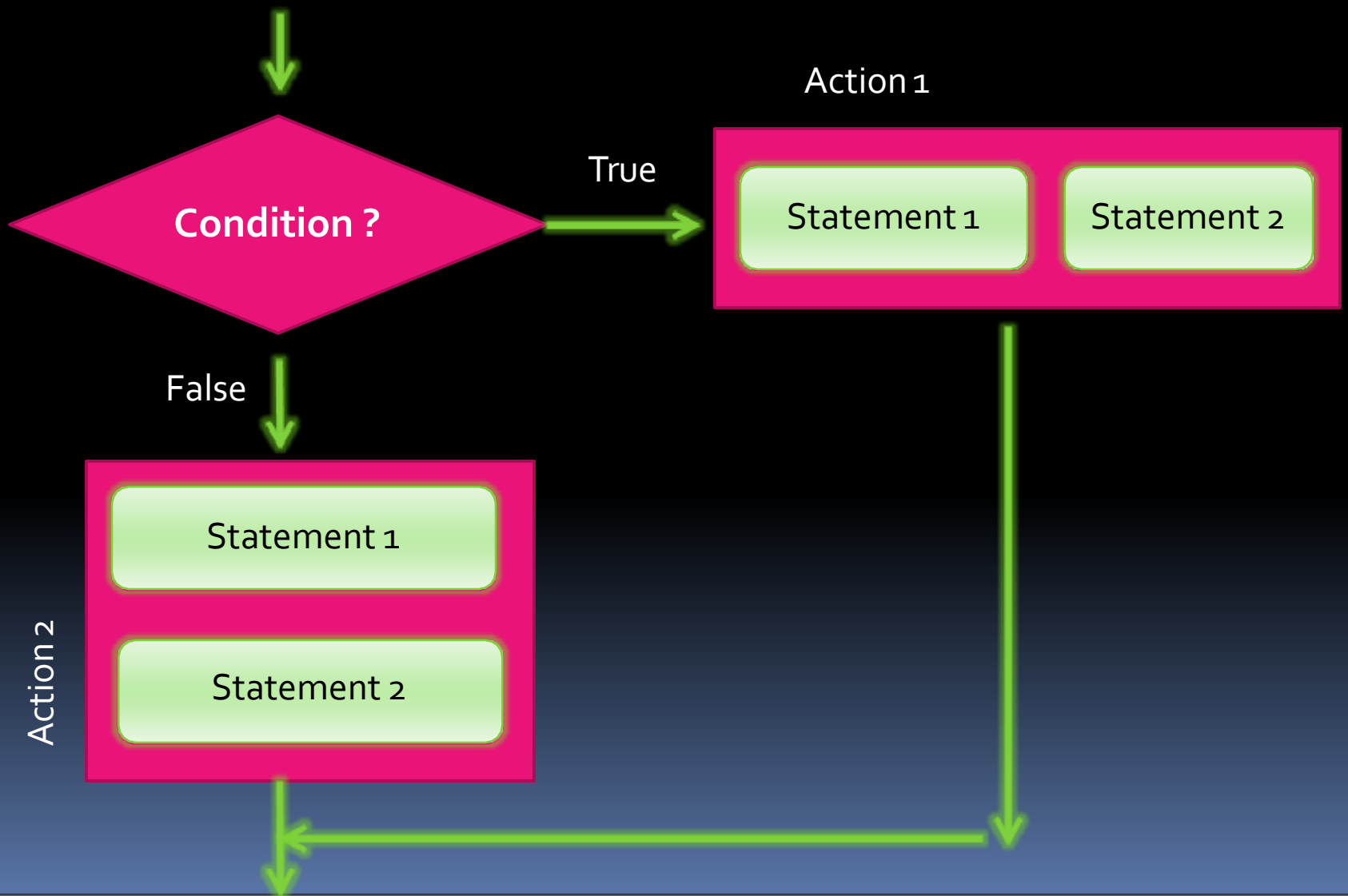
Sequence

- It means python statements are executed one after another i.e. from the first statement to last statement without any jump.

Selection

- It means execution of statement will depend upon the condition. If the condition is true then it will execute Action 1 otherwise Action 2. In Python we use **if** to perform selection

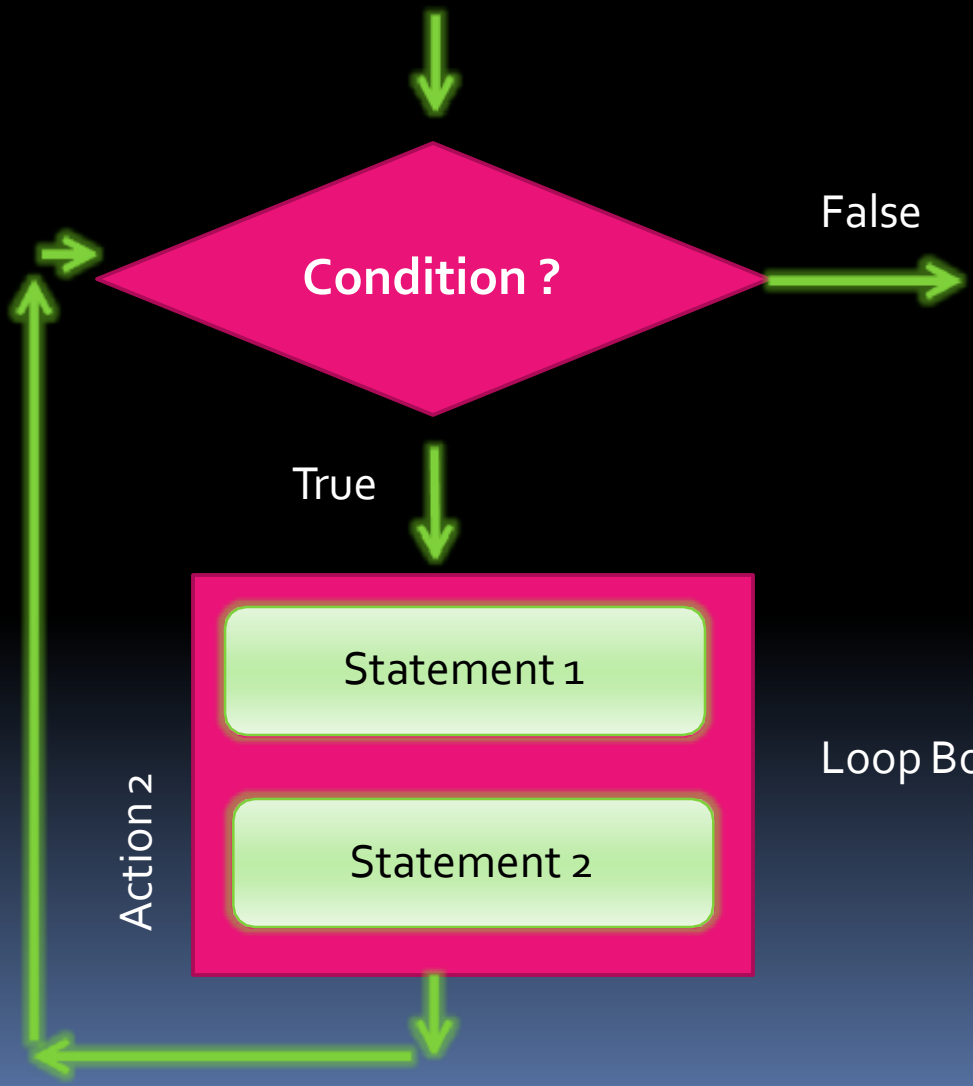
Selection


Condition ?
True
Action 1
Statement 1
Statement 2
False
Action 2
Statement 1
Statement 2

Iteration - Looping

- Iteration means repeating of statement depending upon the condition. Till the time a condition is **True** statements are repeated again and again. Iteration construct is also known as **Looping construct**
- In real world we are using Looping construct like learning table of any number we multiply same number from 1 to 10, Washing clothes using washing machine till the time is over, our school hours starts from assembly to last period, etc.

Iteration - Looping



Python supports **for** and **while** statement for Looping construct

if Statement of Python

- 'if' statement of python is used to execute statements based on condition. It tests the condition and if the condition is true it perform certain action, we can also provide action for false situation.
- if statement in Python is of many forms:
  - if without else statement
  - if with else
  - if with elif
  - Nested if

Simple "if"

- In the simplest form if statement in Python checks the condition and execute the statement if the condition is true and do nothing if the condition is false.
- **Syntax:**

```
if condition:
        Statement1
        Statements ....
```



Points to remember with “if”

- It must contain valid condition which evaluates to either True or False
- Condition must followed by Colon (:) , it is mandatory
- Statement inside if must be at same indentation level.

Example - 1

Input monthly sale of employee and give bonus of 10% if sale is more than 50000 otherwise bonus will be 0

```
bonus = 0
sale = int(input("Enter Monthly Sales :"))
if sale>50000:
        bonus=sale * 10 /100
print("Bonus = " , bonus)
```

To Do..

1. WAP to enter 2 number and print the largest number
2. WAP to enter 3 number and print the largest number

To Do - 1

1. WAP to enter 2 number and print the largest number

```
n1 = int(input("Enter first number "))
n2 = int(input("Enter second number "))
large = n1
if (large<n2):
        large=n2
print("Largest number is ", large)
```

To Do - 2

2. WAP to enter 3 number and print the largest number

```
n1 = int(input("Enter first number "))
n2 = int(input("Enter second number "))
n3 = int(input("Enter third number "))
large = n1
if (large<n2):
        large=n2
if(large<n3):
        large=n3
print("Largest number is ", large)
```

if with else

- **if with else** is used to test the condition and if the condition is **True it perform certain action** and **alternate course of action if the condition is false**.
- **Syntax:**

```
if condition:
        Statements
else:
        Statements
```

Example - 2

Input Age of person and print whether the person is eligible for voting or not

```
age = int(input("Enter your age "))
if age>=18:
        print("Congratulation! you are eligible for voting ")
else:
        print("Sorry! You are not eligible for voting")
```

To Do…

1. **WAP to enter any number and check it is even or odd**
2. WAP to enter any age and check it is teenager or not
3. **WAP to enter monthly sale of Salesman and give him commission i.e. if the monthly sale is more than 500000 then commision will be 10% of monthly sale otherwise 5%**
4. WAP to input any year and check it is Leap Year or Not
5. **WAP to Input any number and print Absolute value of that number**
6. **WAP to input any number and check it is positive or negative number**

if with elif

- if with elif is used where multiple chain of condition is to be checked. Each elif must be followed by condition: and then statement for it. After every elif we can give else which will be executed if all the condition evaluates to false
- Syntax:

```
if condition:
        Statements
elif condition:
        Statements
elif condition:
        Statements
else:
        Statement
```

Example - 3

Input temperature of water and print its physical state

```
temp = int(input("Enter temperature of water "))
if temp>100:
        print("Gaseous State")
elif temp<0:
        print("Solid State")
else:
        print("Liquid State")
```

To Do…

1. WAP to enter marks of 5 subject and calculate total, percentage and also division.

Percentage	Division
>=60	First
>=45	Second
>=33	Third
otherwise	Failed

2. WAP to enter any number and check it is positive, negative or zero number

3.WAP to enter Total Bill amount and calculate discount as per given table and also calculate Net payable amount (total bill – discount)

Total Bill	Discount
>=20000	15% of Total Bill
>=15000	10% of Total Bill
>=10000	5% of Total Bill
otherwise	0 of Total Bill

Nested if

- In this type of "if" we put if within another if as a statement of it. Mostly used in a situation where we want different else for each condition. Syntax:

```
if condition1:
        if condition2:
                statements
        else:
                statements
elif condition3:
        statements
else:
        statements
```

Program to read three numbers and prints them in ascending order

```
x = int(input("Enter first number "))
y = int(input("Enter second number "))
z = int(input("Enter third number "))
if y>=x<=z:
        if y<=z:
                min,mid,max=x,y,z
        else:
                min,mid,max=x,z,y
elif x>=y<=z:
        if x<=z:
                min,mid,max=y,x,z
        else:
                min,mid,max=y,z,x
elif x>=z<=y:
        if x<=y:
                min,mid,max=z,x,y
        else:
                min,mid,max=z,y,x
print("Numbers in ascending order = ",min,mid,max)
```